IMMUNITY

DR.KABSHA

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❖ Innate immunity

The innate immune system is essentially made up of barriers that aim to keep viruses, bacteria, parasites, and other foreign particles out of your body or limit their ability to spread and move throughout the body.

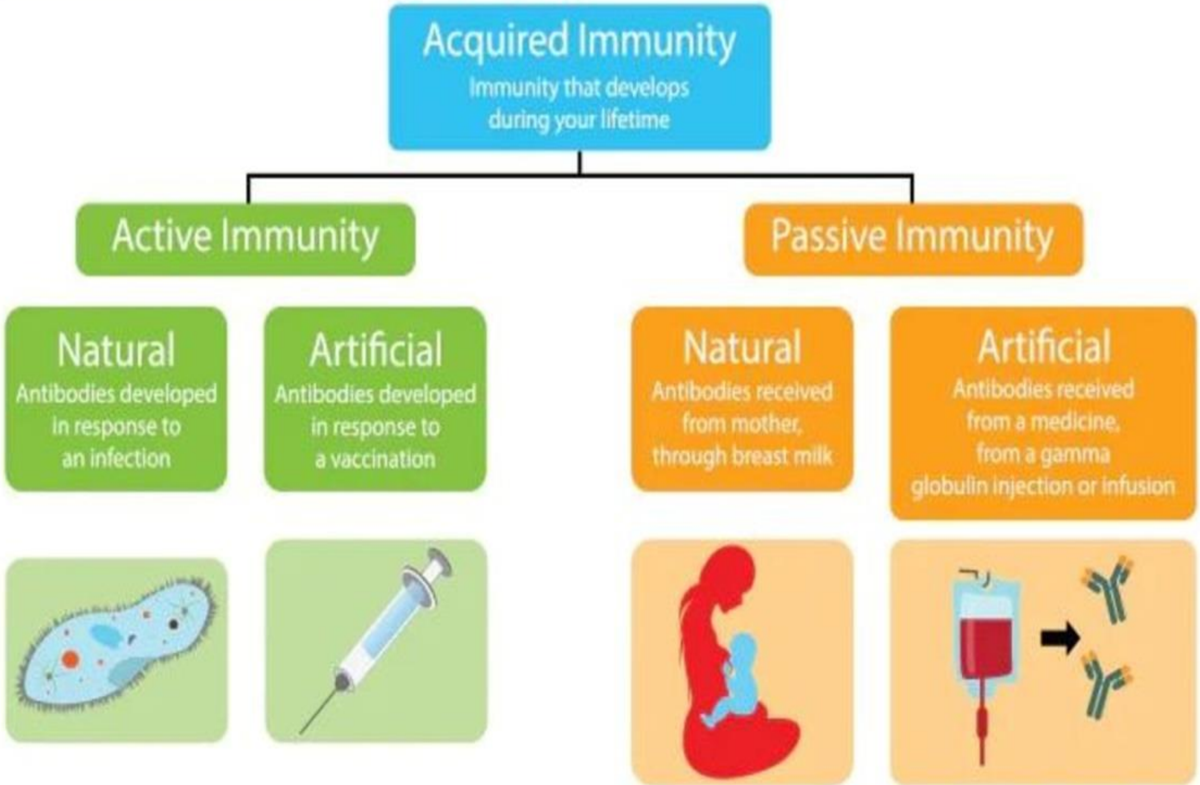
❖ Adaptive immunity

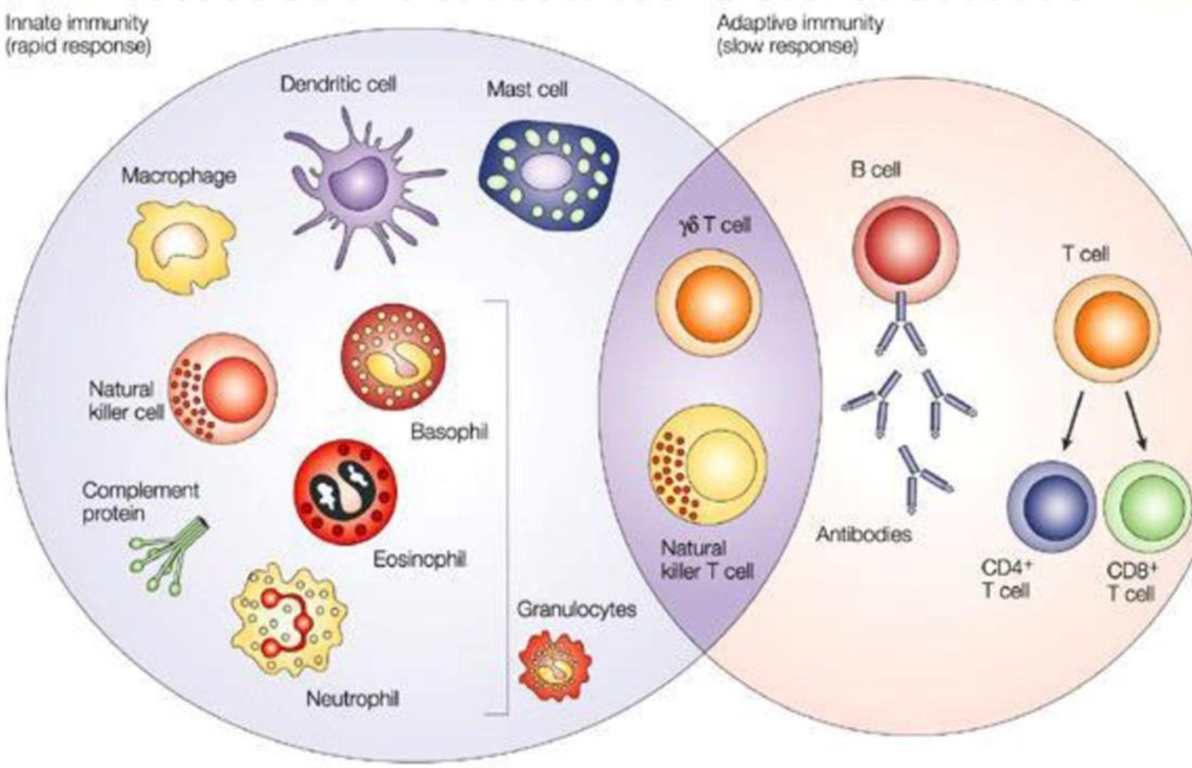
Adaptive immunity (Active or Passive)

(Acquired) immunity a specific defense mechanisms effective against specific microorganism.

- **Depends upon having previous contact with a particular microorganism (memory).**
- **It is mediated either through antibodies (humeral) or cytotoxic effects (cellular)**

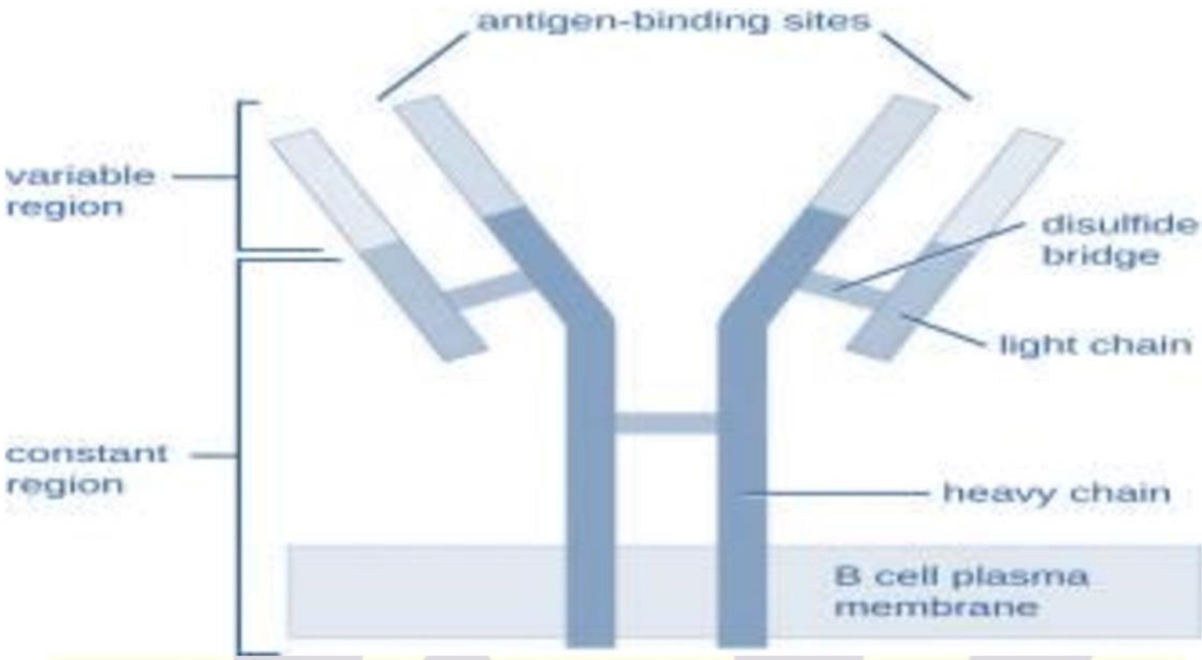
T-lymph	**→ intercellular** **the lymphocytes mature and become specialized T-cells in your thymus**
B-lymphocytes	**→ extracellular**

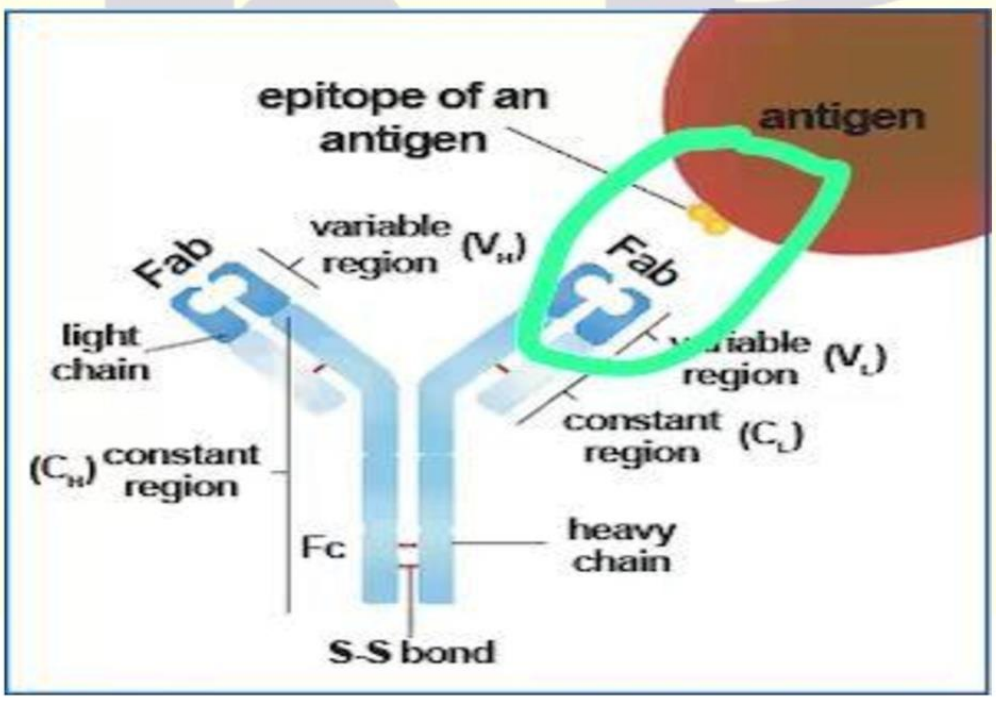

Acquired Immunity
Immunity that develops during your lifetime
Active Immunity
Passive Immunity
Natural
Antibodies developed in response to an infection
Artificial
Antibodies developed in response to a vaccination
Natural
Antibodies received from mother, through breast milk
Artificial
Antibodies received from a medicine, from a gamma globulin injection or infusion


Innate immunity
(rapid response)
Adaptive immunity
(slow response)
Dendritic cell
Mast cell
Macrophage
γδ T cell
B cell
T cell
Natural killer cell
Basophil
Complement protein
Eosinophil
Natural killer T cell
Antibodies
CD4+ T cell
CD8+ T cell
Granulocytes
Neutrophil

1st line defense	2nd line defense
• **Anatomical barrier Skin Cilia in respiratory tract**	• Inflammation • Fever • **Natural killer cells (NK)**
• **Mucous membrane andtheir secretions**	• Phagocytic white blood cells • Antimicrobial substances
• **Bacterial flora**	• Coagulation system • Lactoferrin and transferring **Lysozymes** • Interferons

	IgG	IgA	IgM	IgD	IgE
	Cross placenta to give protection for baby Smallest	Secretory IgA Milk (**colostrum**) ,mucus ,salive	Macroglobin pentameric Largest		**Type I hypersensitivity** **In parasite infection**
	Gama globulin	Monomer in serum Dimer in secretion	1st appear in new infection(modern) fatest	Monomer Differentiation of lymphocyte	
Serum conc	Highest				Lowest
T1/2	**23 day**				


antigen-binding sites
variable region
constant region
disulfide bridge
light chain
heavy chain
B cell plasma membrane


epitope of an antigen
antigen
Fab
variable region (V_H)
Fab
light chain
variable region (V_L)
constant region (C_L)
(C_H) constant region
Fc
heavy chain
S-S bond

Epitope

"Antigenic determinant" Part of an antigen that is recognized by the immune system. Specially by antibodies, B cell, T cell

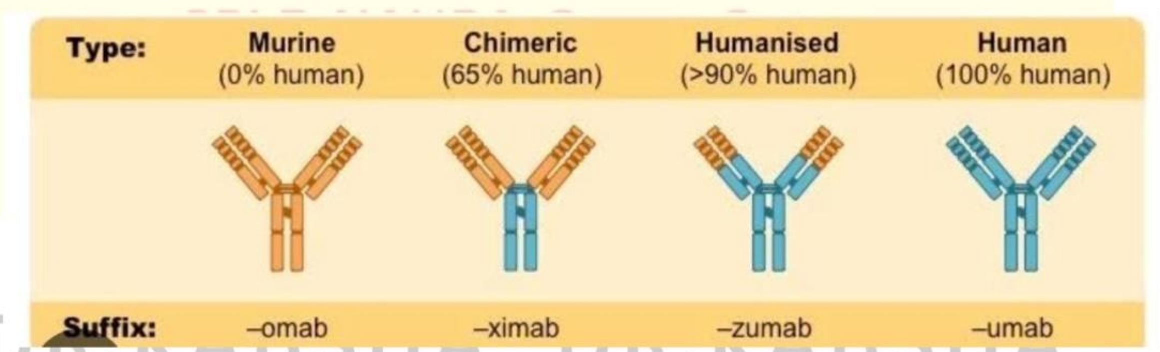
Anergy

A term in immunobiology that describes a lack of reaction by the body's defense mechanisms to foreign substances

Hapten	• Small molecule which could never induce an immuneresponse when administered alone but can induce immune response when coupled to a carrier protein
Antigen	• Foreign material causing an immune response
Antibody	• Specific substance formed in the body in response to antigenic stimulation and react specifically with antigen

Hybridoma?

- Monoclonal antibody



❖ DEFINITATIONS

Chemo taxis

The migration of cells to an inflammatory site following interaction with chemo attractants or chemokine's.

→ **EX. C_5a of the complement system.**

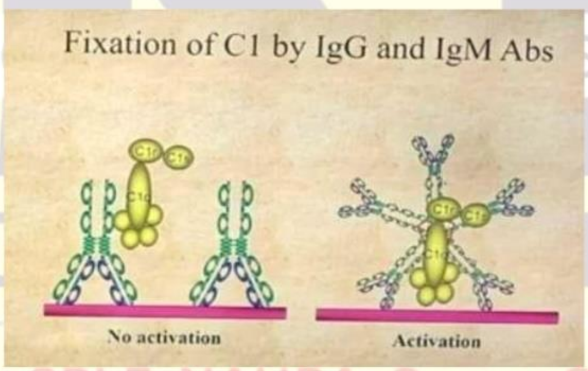
Opsonization

The process by which plasma proteins are attached to a foreign substance and prepare it for phagocytosis.

Epitope (antigenic determinant)

Small part of the immunogenic that is recognized by B or T cells.

→ **Can be as few as 6 amino acids**

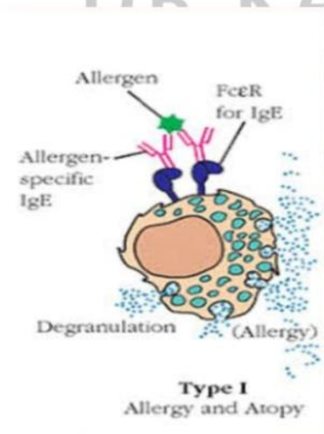
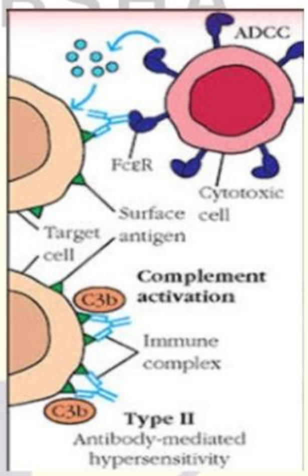
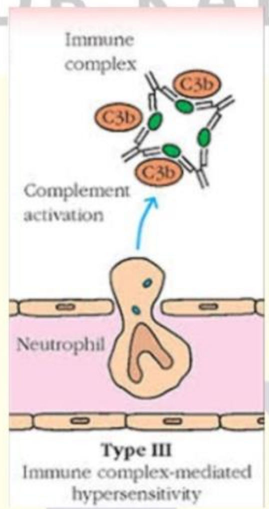
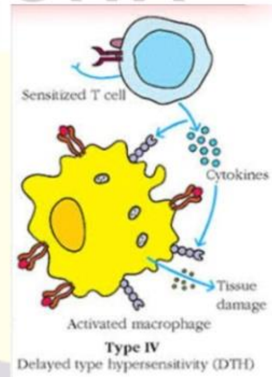


Which of the following immunoglobulin activates complement system?

A) IgM
B) IgE
C) IgA
D) IgG

Which of the following represents the immunity acquired by give livingorganism with attenuated virus?

A. Local immunity
B. Passive immunity
C. Natural active immune
D. Artificial active immunity

Type I IgE immediate	Type II IgG/ IgM cytotoxic	Type III IgG/ IgM Immune complex	Type IV cell mediated delayed
 **Mast cells** ○ Anaphylaxis , ○ Allergies →e.g. peanut ○ Asthma ○ ??? ○ ???	 **Macrophages** ○ Hemolytic anemia ○ ErythroBlastosis fetalis ○ Blood transfusion	 **Neutrophil** ○ Lupus ○ serum sickness ○ RA ○ glomerulonephritis	 **T cells - Macrophages** ○ T.B ○ contact dermatitis ○ graft rejection

❖ **QUESTIONS**

Which cell has antibody on its surface?

A. **IgE —>hypersensitivity (allergic reaction)**
B. **IgA —-> present in milk.**
C. **IgM —-> primarily immune response.**

Longest half-life?

- **IgG**

Frist Ig antibody appears?

- **IgM**

What's the immunoglobulin in MALT (mucosa-associated lymphatic system)?

- **IgA**

New infection?

- **Lgm**

Hypersinsitivelg fond in fat tissue?

- **IgE**

Frist release in body?

- **IgM**

Ig in milk?

- **IgA**

Ig in serum,saliva and Git ?

- IgA

Ig in asthma?

- IGE

Angioedama type of hypersensitive?

- LgE

Peanut allergy symptom?

- **Type 1(IgE)**

Strongest IG?

- IgG

Fastest IG?

- IgM

Immunoglobulin found in serum predominantly?

- IgG

Ig on the blood?

- IgG

Ig has gamma globulin?

- **IgG**

Penicillin allergy?

- **IgE**

Immunosupressant cause hairsutism?

- **Cyclosporine**

By which one antibody is produced?

- **B lymphocytes**

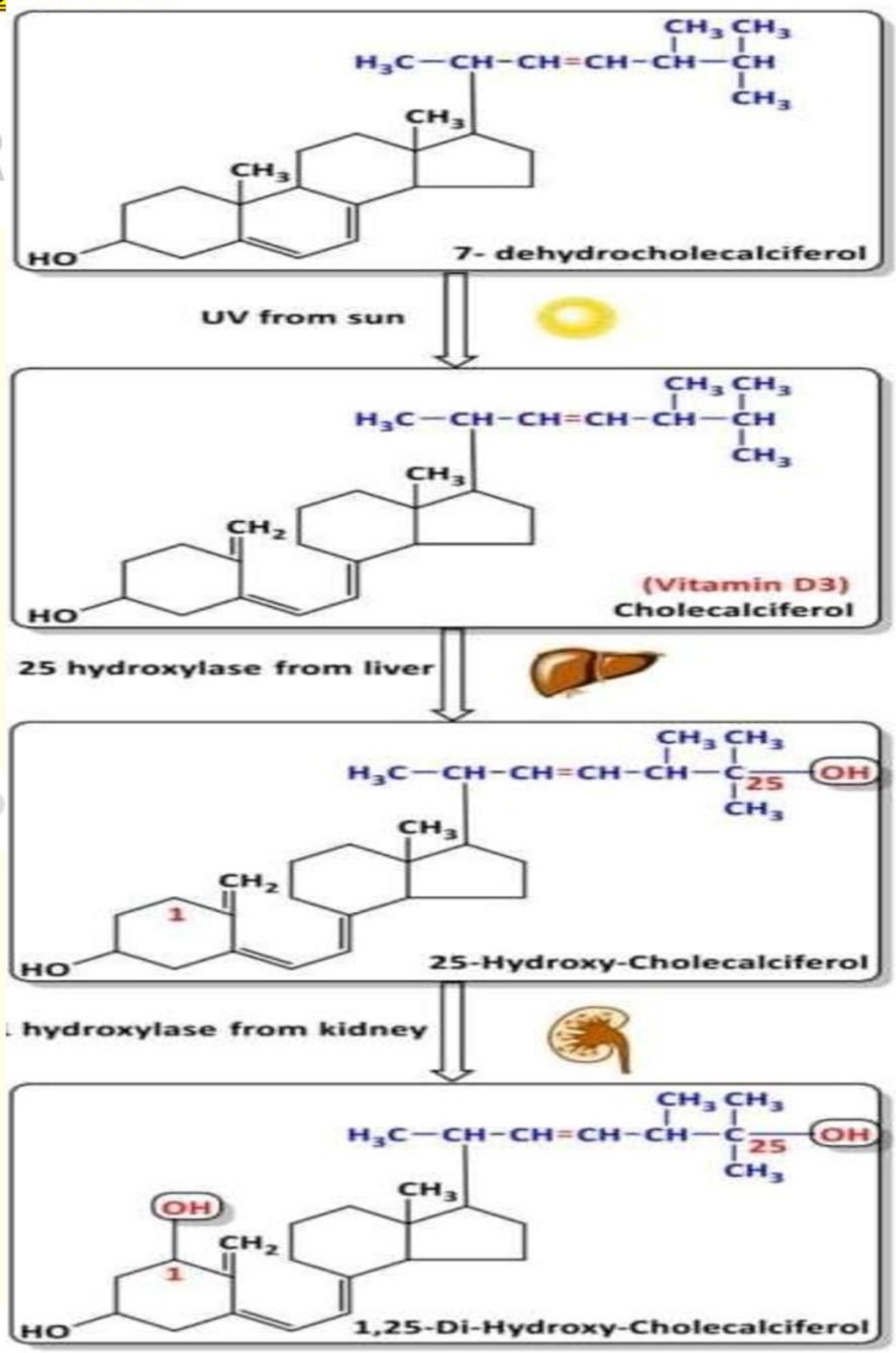
Immunoglobulin found in parasite infection?

- **IgE**

❖VITAMINS

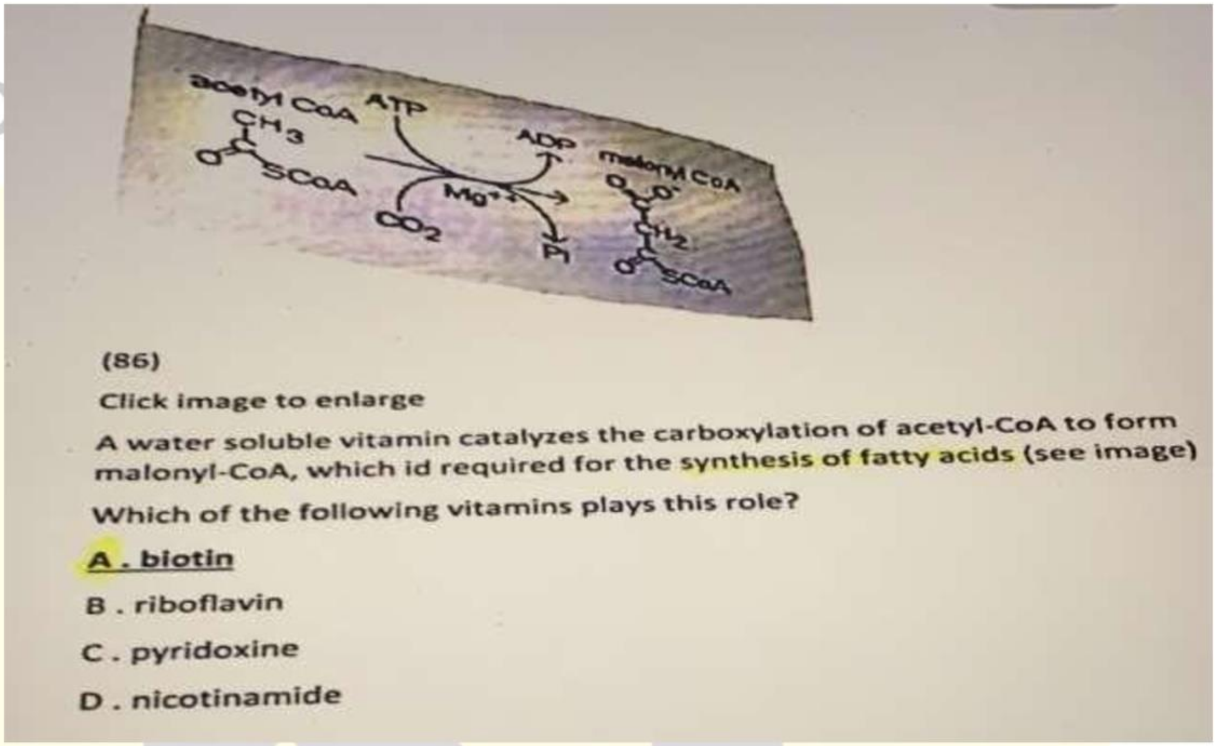
	Adult	Pregnant	Breast feeding	Generic
Ca^{+2}	1000 mg/day	1200 mg/day	-	1200 mg/day
Vitamin D	600 IU	600 IU	600 IU	600 IU 800 IU→ above 70
Vitamin C	♂ 90 MG ♀ 75 MG	-	-	-
Vitamin A	♂ 1000 MCG ♀ 800G	900 Mcg	1200 mcg	-
Folic acid	♂ 400 MCG ♀ 400-800 MCG	600 mcg	500 mcg	-
$FeSO_5$	1200 mg/day (1*1) 325 mg/day(1*3)	-	-	-
Elemental Fe	150 mg/day (1*1) 50 mg/day (1*3)	-	-	-

Vit	*Other name*	*Function*	*Deficiency*	
A	**Retinol** **B -carotene**	**Vision**	**Night blindness, xerophalmia**	**C.I in pregnancy with high doses (15000IU)**
D	**D2 ergocalciferolD3 cholecalciferol**	**Bone health & abs ca&….** **↓ Breast cancer risk**	**Osteomlasia , cvs**	
E	**Tocopherol**	**Spermatogenesis (fertility)**	**Sickle cell disease**	
K	**Phylloquinone** **Antidote of warfarin**	**Blood clotting**	**bleeding**	


7- dehydrocholecalciferol
UV from sun
(Vitamin D3)
Cholecalciferol
25 hydroxylase from liver
25-Hydroxy-Cholecalciferol
hydroxylase from kidney
1,25-Di-Hydroxy-Cholecalciferol

Calcitriol

Vit	Other name	Function	Deficiency
B1	**Thiamin**	CHO metabolism through cofactors	**Beri-Beri**
B2	**Riboflavin**	Energy metabolism	**Riboflavinosis Cracks and sores inmouth and lips → Stomatitis**
B3	***Niacin***	**↓ cholesterol& LDL** ↑↑↑ **HDL** **By high dose**	○ **Pellagra** **Vasodilator (VD)** ○ **Niacin flush caused transient increase in liver enzymes**
B5	**Pantothenic**	CHO, protein, fat metabolism	
B6	**Pyridoxine**	○ Treat pregnancy vomiting Antidote for INH	**Microcytic hypochromic anemia Neuropathy**
B7	○ **Biotin**	Beneficial in skin, hair and nails	
B9	Folic acid take before pregnancy	Synthesis of nucleic acid • Formation of blood cells • Activates B12	Megaloblastic anemia • **Neural tube defect (spina bfida)** • **Recommended in pregnancy**
B12	**Cobalamin**	B12 test? schilling Absorbed by intrinsic factor Drug effect on abs?	**Pernicious anemia**
C	**Ascorbic acid**	***Inahnce abs of iron** ***Used In cold (severity) Zn in duration** *** Production of collagen** *** Preservative in preparation**	**Scurvy**



(86)

Click image to enlarge

A water soluble vitamin catalyzes the carboxylation of acetyl-CoA to form malonyl-CoA, which id required for the synthesis of fatty acids (see image)

Which of the following vitamins plays this role?

A . biotin

B . riboflavin

C . pyridoxine

D . nicotinamide

❖ Minerals

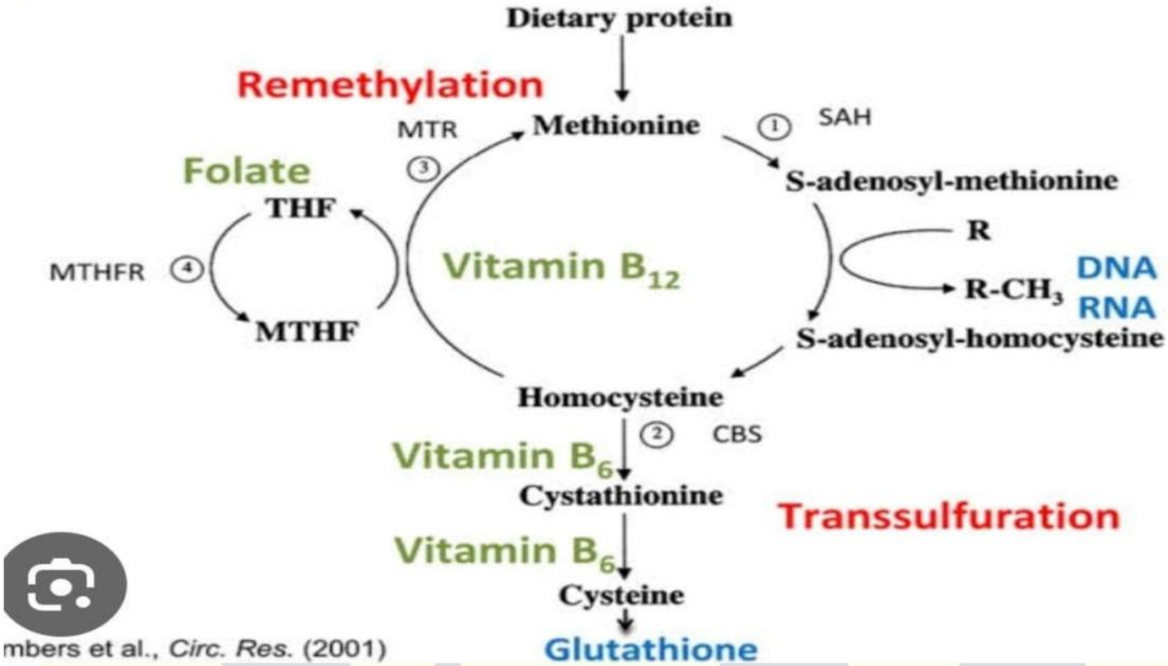
1. Zn?
2. Cr?
3. Selenium?
4. Iodine?
   - Goiter

Copper is essential in iron abs in gut

Ca &vitamin D>>Pt with colitis take cortisone, osteoporosis

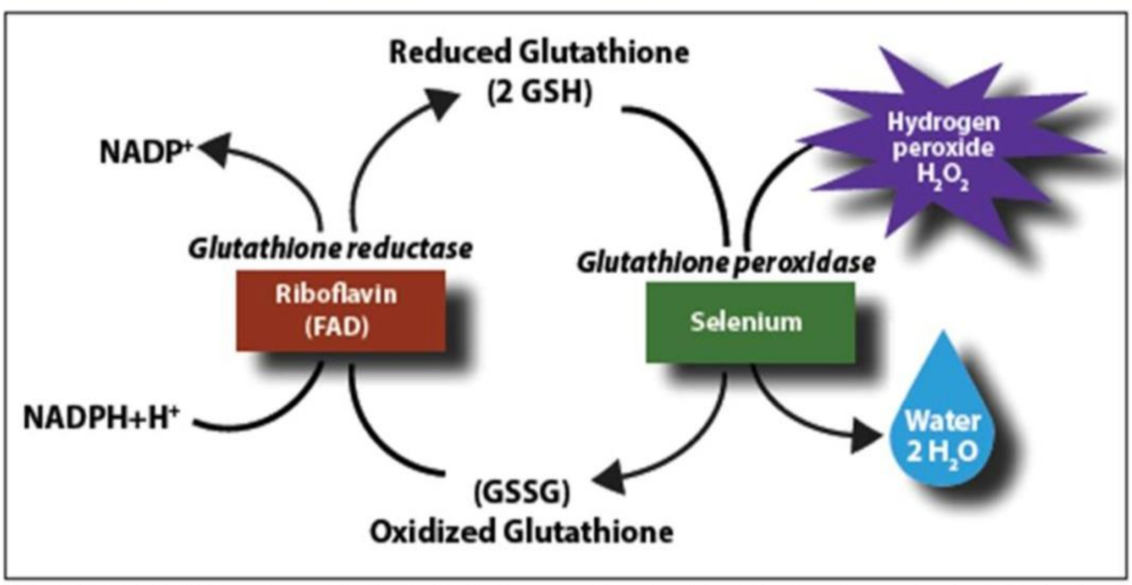
Which one within mitochondria?

- Co enzyme Q10 with selenium



Supplement protect from cardiomyopathy after bariatric surgery?

- **Selenium**



Drug categories in pregnancy are

Category A	○ no risk on fetus
Category B	○ no risk human but risk on animal ○ No study on human & no risk on animal
Category C	○ risk can't be ruled out ○ No study on human but +ve risk on animal
Category D	○ positive evidence of risk
Category X	○ Contraindicated in pregnancy(statin ,methotrexate ,warfarin, Finasteride)

According to FDA categories of drug safety during pregnancy amoxicillin is under category B what does that mean

A) No adequate animal or human study
B) Controlled human study show no fetal risk
C) Evidence of human fetal risk exist but benefits may outweigh
D) Animal studies show no risk to the fetus and no controlled human studies

Nicotine

Decrease blood flow to uterine

ACE-I

contraindicated during the second and third trimesters of pregnancy because of increased risk of fetal renal damage

.Warfarin

Nasal bone hypoplasia in neonate

Finasteride

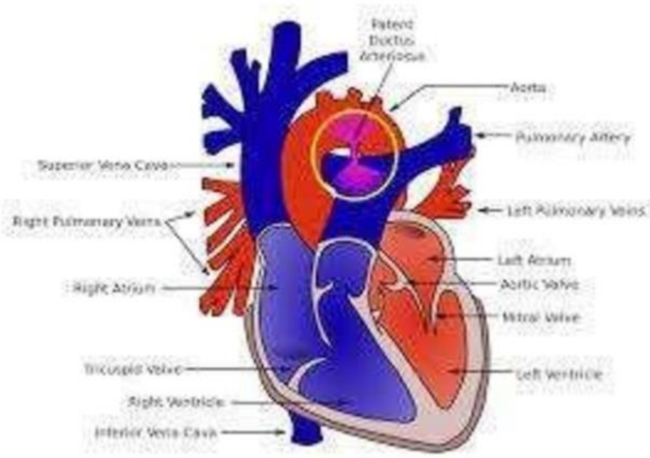
Genital malformation in the infant

Castor oil

Is contraindicated as laxative

Nitrofurantoin?

NSIAD?



(PDA)

Is a persistent opening between the two major blood vessels leading from the heart .The opening (ductus arteriosus) is a normal part of a baby's circulatory system in the womb that usually closes shortly after birth. If it remains open, it's called a patent ductus arteriosus. (PDA)

- **IV indomethacin ttt PDA in preterm**
- **Alprostadil (PG E1) keep ductus arteriosus open**

❖ Use in pregnancy

- **Zidovudine →Antiviral give pregnant:**
- **Zidovudine administered to mildly symptomatic women with HIV infection in the antepartum (100mg orally 5 times/day), intrapartum (2 mg/kg intravenously over 1 hour then 1 mg/kg/h) and then to the neonate for 6 weeks (2 mg/kg), significantly reduced the rate of vertical HIV transmission**
- **Acyclovir?**
- **Oseltamivir : ttt of H1N1 influenza A virus**

❖ Abbreviation

AC • Before meal	
PC • After meal	
HS - BT • At bedtime	
PO • By oral	
PR • By rectum (fattest roat)	
PRN • As needed (Dangerous abbreviation)	

Gtt drop	
Ou→both eye	AU→both ear
Od→right eye	AD→ right ear
O.S →left eye	A.S→ left ear
Tsp. = 5ml teaspoonful	tbsp. = 15ml table spoonful

OD →Once a day	Q→ every
BID →Twice a day	QD→ every day
TID →Three a day	QOD →Every other day
QID →Four times a day	QH → Every hour
C →With	QAM →Every morning
S →Without	QPM → Every night

Q4H →every 4 hours

IVP→ intravenous push LB= pound mEq = mill equivalent

IVPB→ intravenous piggyback?

- Vincristine, vinblastine, vincristine For IV infusion administration only , if give Intrathecal cause death

PH Stomach :1.5 to 3.5	2-8 c refrigerator cold
PH Urine :4.5 to 8	8-25 c cool
PH Saliva: 6.2-7.6	25-30 c room temperature
PH intestine: 5.7-7.4	30-40 c warm temperature

❖ SOAP note

The SOAP note (an acronym for subjective, objective, assessment, and plan) is a method of documentation employed by healthcare providers to write out notes in a patient's chart, along with other common formats

⇒S = Subjective findings

The patient specific subjective information that gives a basis for the recognition of a pharmacotherapy problem or indication for pharmacist intervention.

EX: Chief complaint and duration or severity of symptoms

include

- **Patient complaints.**
- **Symptoms.**
- **Observations of the medical team**

⇒O = Objective findings

The patient specific objective nformations that gives a basis for the recognition of a pharmacotherapy problem or indication for pharmacist intervention.

→Clinical signs.

- **Results of laboratory tests.**
- **Findings from an examination.**
- **Results of procedures**

EX: X –Ray, ECG, and EEG etc.

⇒A = Assessment

Pharmacist evaluation of the subjective and objective findings.

→ include

- **Interpretation of the subjective and objective data.**
- **With pharmaceutical knowledge and existing evidence base.**
- **Develop and document a therapeutic plan.**
- **Follow the response to therapy.**
- **Develop a systemic process for assessing problems so that assessment is complete and accurate.**
- **References.**

⇒P = Plan

- **The plan includes recommended actions when appropriate.**
- **The plan is usually expanded to describe information included in the monitoring and follow ups**

❖Questions

Pt comes to ER and he told to doctor I feel, headache this is

A) S
B) O
C) A
D) P

The doctor fined the blood pressure of patient is high blood pressure … this is

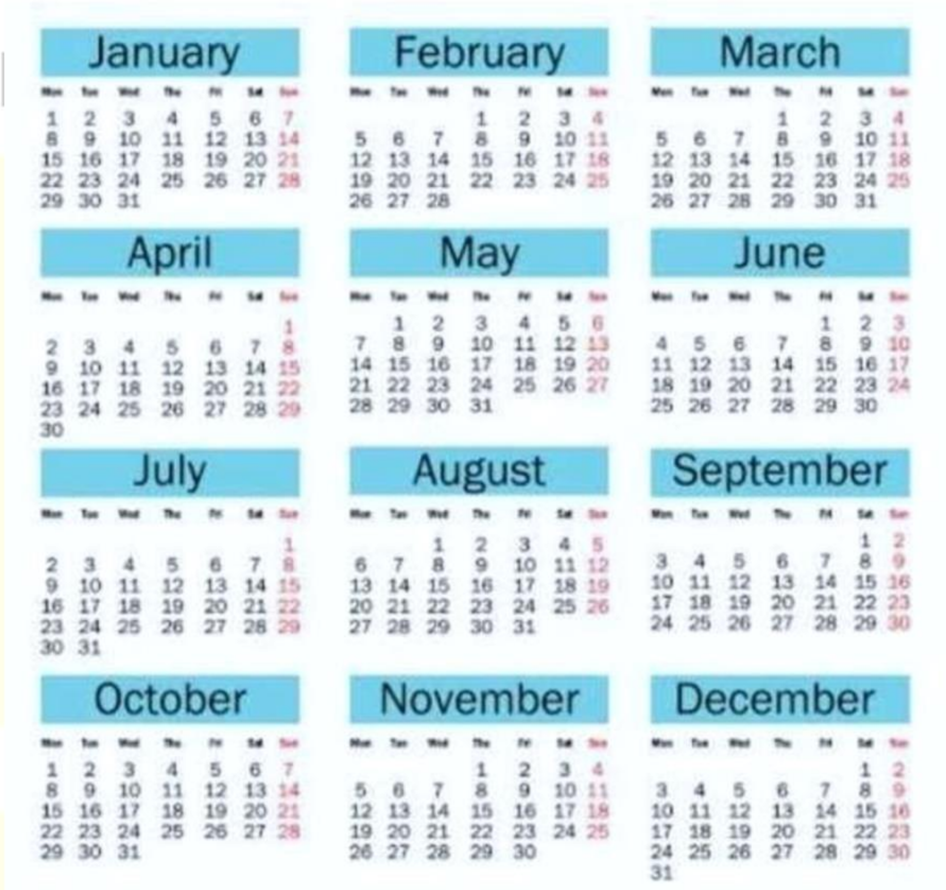
A) S
B) O
C) A
D) P

Problem list: Hypertension, Coronary artery disease…..This is

A) S
B) O
C) A
D) P

Doctor said continue on your medication and follow the lab every 3 month…this is

A) S
B) O
C) A
D) P



Ap SJN 30 days Feb 28

Q- Medications expire in 2020 DEC when last time takes medication?

A) 30 Dec 2020

B) 31 Dec 2020

C) 30 NOV 2020

❖ Anemia classification

→Normal MCV 80-100: normocytic anemia eg: CKD, malignancy, acute blood loss, bone marrow failure

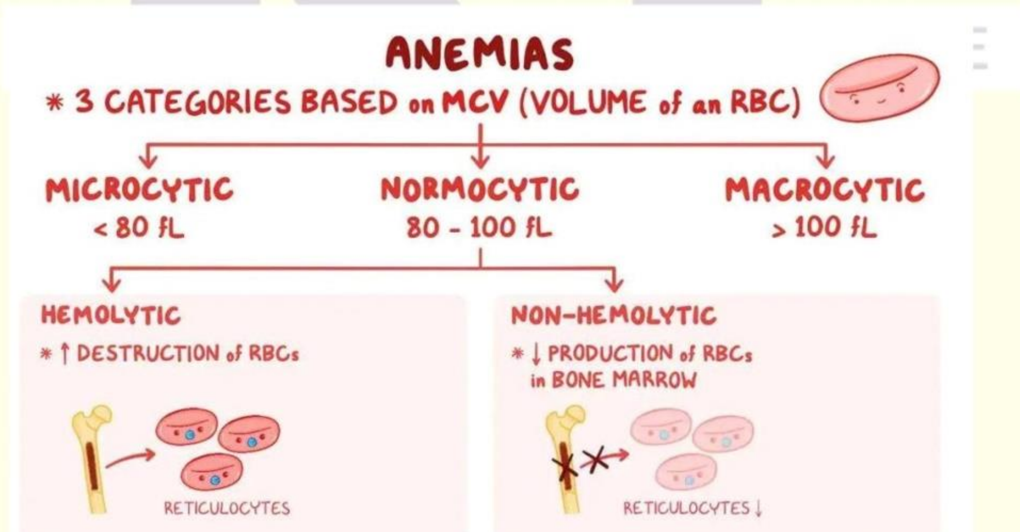
→Low <80: microcytic: Iron deficiency (hypochromic)

→High> 100: macro: b12 and foliate deficiency (hyper chromic)

→Megaloblastic anemia, Macrocytic anemia: {Folic (Vit B9), Vit B12}

→Pernicious anemia: {Decrease Vit B12}

Iron →black stool



❖Anemia

Low <80: microcytic: Iron deficiency

- ⇒A. Iron deficiency anemia(Decrease iron)

- Iron deficiency anemia: (**Decrease iron)**
- § **Oral iron**: {ferrous sulfate, ferrous fumarate}. SE:

GI, constipation, **dark stool**.

Copper & vit c: essential for iron absorption . #

DI: decrees levothyroxine, levodopa, methyldopa /

PPIs: decrease iron

\# **Dose**: 325 mg TID {elemental iron = 65 mg}.

\# **Antidote**: deferoxamine {non-receptor mechanism, because it is bind to free iron} §

Parenteral iron: {iron dextran, iron sucrose} # Parenteral iron is restricted to: unable to tolerate oral iron, extensive CKD

\# Black box warning: anaphylactic shock

Iron store in the body: Hemosiderin & ferritin ?

- ⇒B.Normocytic anemia{**Decrease Erythropoietin (EPO)}**
- Erythropoiesis stimulating agent (ESA): Epoetin alfa, Epoetin beta, Darbepoetin
- Iron is important for ESA to be effective
- risk of death, serious cardiovascular events and stroke
  When Hgb level **> 11 g/dl**

⇒C. Aplastic anemia

{Bone marrow fails to make RBCs} Immunosuppressant, blood transfusion, bone marrow transplantation

⇒D. Hemolytic anemia (sickle cell disease, G6PD deficiency)

{RBCs destroyed before their lifespan}

Glucose-6-phosphate dehydrogenase (G6PD) deficiencyan X-linked inherited disorder that most affects persons of African, Asian, M editerranean or Middle Eastern descent. The G6PD enzym e protects RBCs from harm ful

substances (e.g., reactive oxygen species). W ithout sufficient levels of G6PD, RBCs hem olyze (break apart) 24 - 72 hours

after exposure to oxidative stress

⇒F. Megaloblastic anemia, Macrocytic anemia

{↓ Folic (Vit B9), ↓ Vit B12}

- **⇒G.Pernicious anemia: {Decrease Vit B12}**

- **Using Schilling** test to detect amount of Vit B12 # Absorbed by **intrinsic factor**
- {* if there is a lack of intrinsic factor à lead to decrease Vit B12 absorption à Cause Precious anemia}
- Common medication decreases Vit B12؟؟؟ {Metformin, PPIs,

❖ Methotrexate in RA

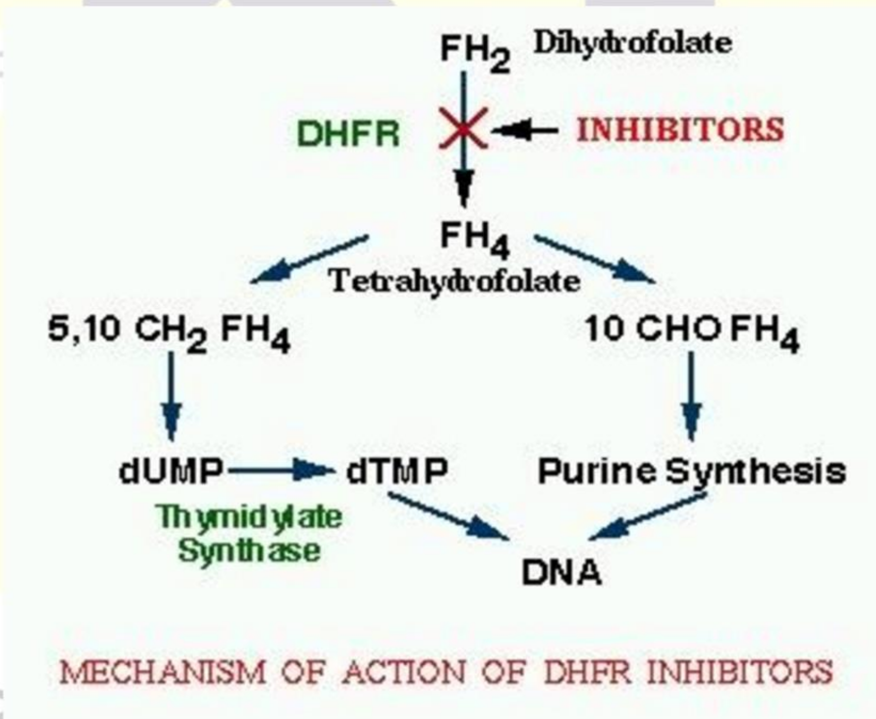
Q) Methotrexate & folic acid?

7.5-15 mg once weekly

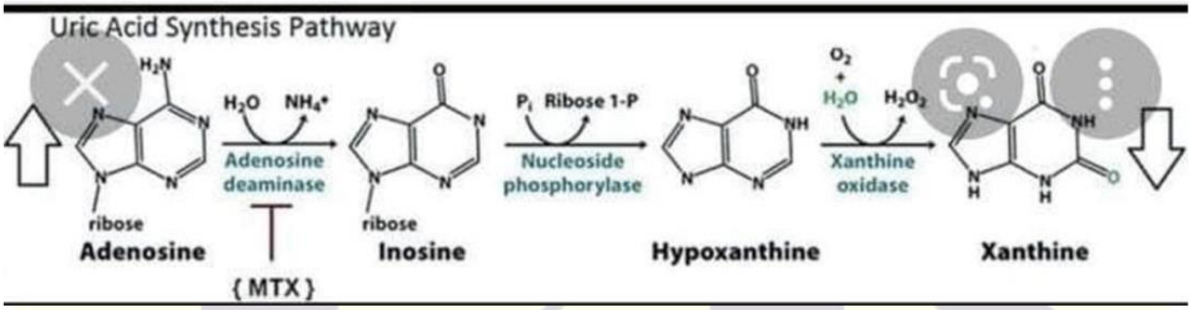
Take folic acid (leucoverin) (B9) not in the same day

⇒Can be used. Intrathecal Use:

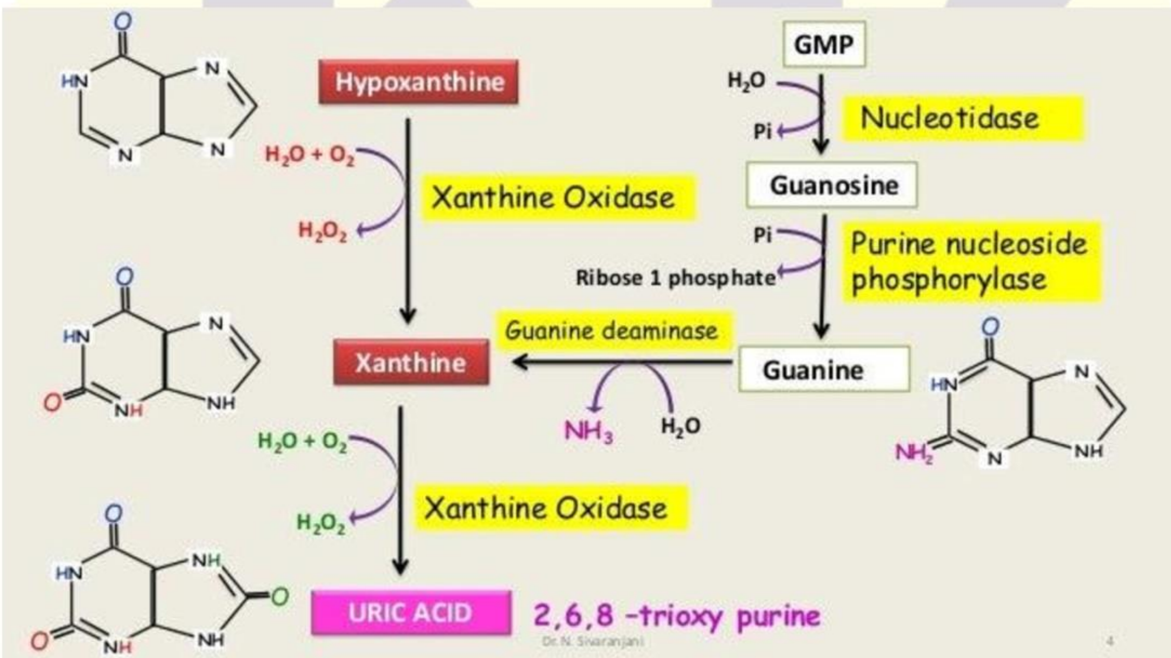
- **With other drug in cancer ,lymphoma ,leukemia**
- **Low dose in autoimmune disease : RA, crohns disease, psoriasis**
- **Category x abortifacient with misoprostol**
- **Contra-indicated in breastfeeding**
- **S/E dose related: hepto & renal toxicity**



Mtx (methotrexate)in cancer is antimetabolite ..inhibits dihydrofolate reductase so inhibit formation of tetrahydrobiopteroate….inhibit folic synthesis that required for DNA synthesis



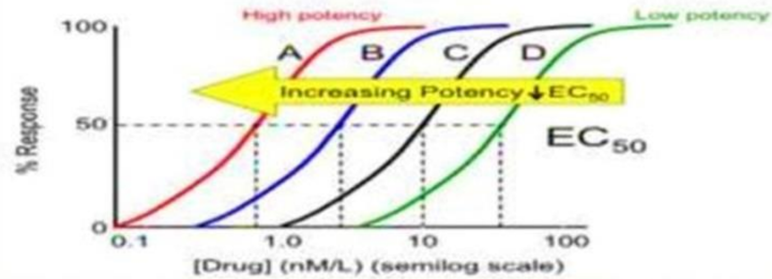
Mtx in RA…inhibition of enzymes involved in purine metabolism, leading to accumulation of adenosine (anti-inflammatory), inhibition of T cell activation



Potency and Efficacy

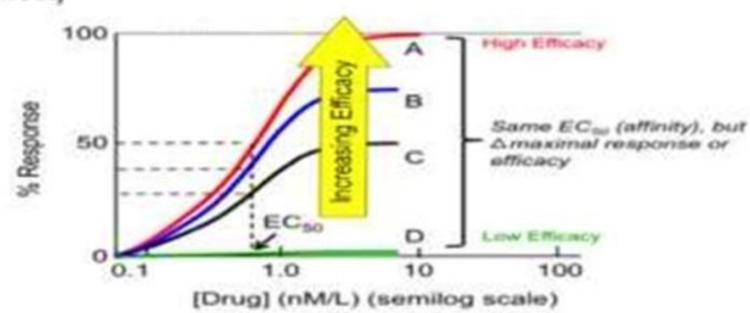
Potency

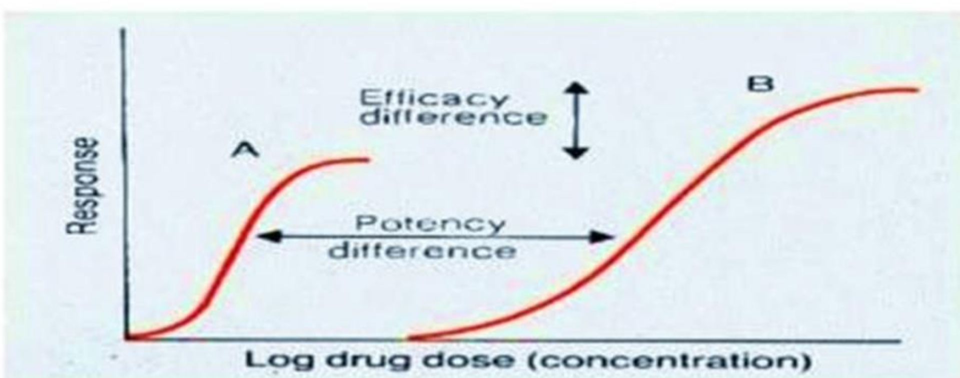
Potency is the concentration (EC_{50}) or dose (ED_{50}) of a drug required to produce 50% of that drug's maximal effect.



Efficacy

Efficacy (E_{max}) is the maximum effect which can be expected from this drug (i.e. when this magnitude of effect is reached, increasing the dose will not produce a greater magnitude of effect)





❖ BISPHOSPHONATES

❖ empty stomach? ……………………???

Alendronate → daily ,, weekly

Ibandronate → monthly

Zelondronic acid →once per a year iv monitor kidney

Risedronate→ daily ,, weekly,, monthly

S/E→Osteonecrosis of the jaw

Alendronate :

Prophylaxisis: 35mg once weekly

treatment : 70mg once weekly or 10mg

daily

esophageal ulcer ?

- alendronate

Gastric ulcer?

- Oral bisphosphonates Risedronate

Oral→Alendronate Ibandronate

DEXA test

T score

Normal: > -1

Osteopenia (low bone mass): -1 to -2.4

Osteoporosis: < -2.5

⇒T.B

❖ Latent > no isolation

INH (9months) or RIF (4months)) or INH+RIF (3months)

❖ Active isolation in negative pressure room because transmission airborne

initial phase → RIPE 2 months

continues phase→ RIF+INH 4 months

Pregnant active TB → Pregnant women should start treatment as soon as TB is suspected. The preferred initial treatment regimen is INH, rifampin (RIF), and ethambutol (EMB) daily for 2 months, followed by INH and RIF daily, or twice weekly for 7 months (for a total of 9 months of treatment

- **Isoniazid (INH)**
- **Rifampin (RIF)**
- **ethambutol (EMB)**
- **Pyrazinamide (PZA)**

❖ Diarrhea

Prophylaxis → bismuth subsalicylate

Mild and moderate → loperamide

Sever → azithromycin 1000mg …. Or …

⇒Fluid

- **<10 kg …100ml / kg**
- **10-20kg …50ml /kg above 10kg +1000ml**
- **20kg >…….. 20 ml / kg above 20kg+1500ml**

Glaucoma!	***Latanoprost (Xalatan)***
Glaucoma + Asthma	*Latanoprost + Betaxolol*
Glaucoma + pregnancy	*Brimonidine or Timolol

When two different eye drop preparations are used at the same time of day, wait for at least five minutes before putting the second drop

❖ Qualitative tests for different types of phytochemicals -

Qualitative Tests for Different Types of Phytochemicals	
Phytochemical Type	**Test**
Alkaloids	Mayer's test
	Wagner's test
	Dragendroff's test
	Hager's test
Carbohydrates	Molish's test
	Fehling's test
	Barfoed's test
	Benedict's test
	Borntrager's test
	Legal's test
Saponins	Froth test
	Foam test
Proteins and amino acids	Millon's test
	Biuret test
	Ninhydrin test
	Xanthoproteic test
Flavonoids	Alkaline reagent test
	Lead acetate test
	Magnesium and hydrochloric acid reduction
Phytosterols	Libermann-Burchard's test
	Salkowski's test
Phenols	Ferric Chloride test
Tannins	Gelatin test
Diterpenes	Copper acetate test
Cardio Glycoside	Killer-Kallani-test
Terpenoids	Salkowski test
Fixed oil and Fats	Sport test
	Saponification Test
Gum and mucilage	Ruthenium red solution

- ***Drangdroffs test***→**detection of alkaloid**
- ***Baljet test***→**detection of glycoside**
- ***Killer kallani***→**detection of cardiac glycoside (digoxin)**

Q) Sport & saponification?

A) **Fixed oil & fat**

………………………………………………………………………??

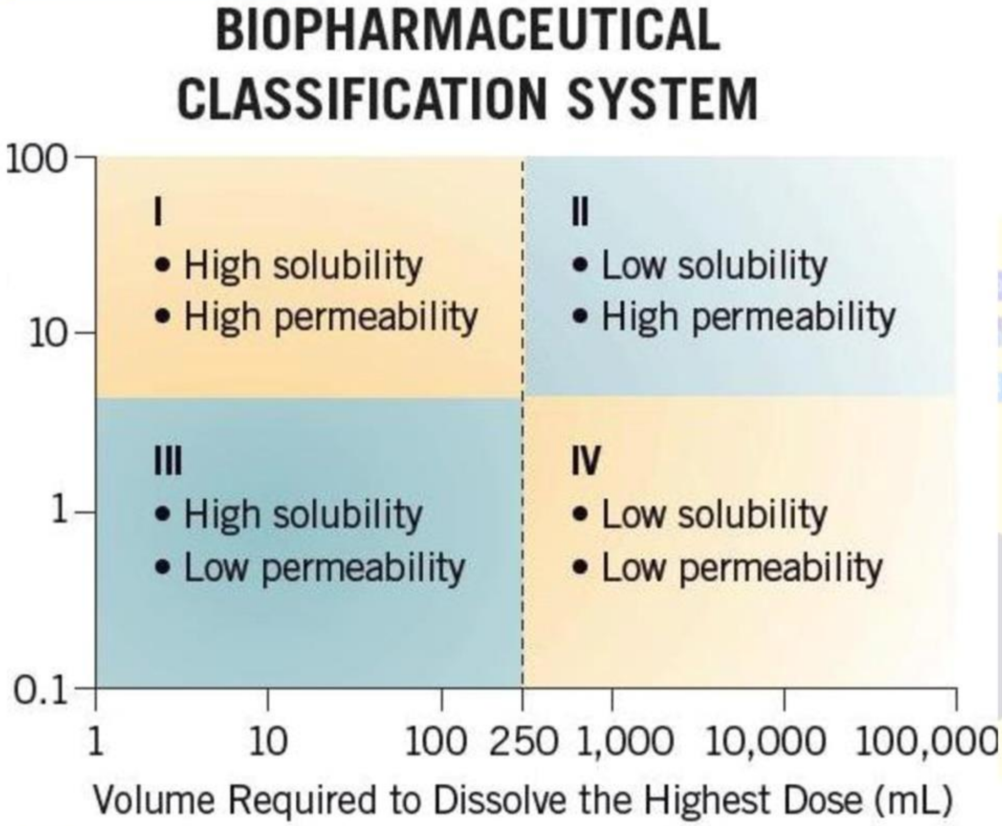
Note

- **Electron transport chain occurred in inner mitochondrial membrane ATP …energy carrier**
- **Aerobic respiration**
- **2 ATP glycolysis**
- **2 ATP Krebs cycle**
- **34 ATP electron transport chain**

Solubility DIFINATION

Solubility is defined as concentration of solute in a saturated solution at a certain temperature

Descriptive terms	Relative amounts of solvents to dissolve 1 part of solute
Very soluble	Less than 1
Freely soluble	From 1-10
Soluble	From 10-30
Sparingly soluble	From 30-100
Slightly soluble	From 100-1000
Very slightly soluble	From 1000-10,000
Insoluble or practically insoluble	More than 10,000


BIOPHARMACEUTICAL
CLASSIFICATION SYSTEM
I
• High solubility
• High permeability
II
• Low solubility
• High permeability
III
• High solubility
• Low permeability
IV
• Low solubility
• Low permeability
100
10
1
0.1
1
10
100
250
1,000
10,000
100,000
Volume Required to Dissolve the Highest Dose (mL)

Q

What is the Latin abbreviation for "after meals?

A) A.C
B) A.A
C) P.C

Which of the following is the abbreviation for drug to be taken by mouth

A) P.O
B) p.R
C) PRN
D) U.D

Which of the following drugs used in rheumatoid arthritics can be dosed once a week?

A) hydroxychloroquine
B) Methotrexate.
C) sulfasalazine
D) prednisone

A 27-year-old man suffers from hepatitis C virus infection. His level of alanine aminotransferase and aspartate aminotransferase are high. In addition, liver biopsy showed mild fibrosis

(See lab results).	Test Result	Normal Values
Alanine aminotransferase	75	5-40 IU/L
Aspartate aminotransferase	55	12-40 IU/L

Which of the following drugs should be avoided in his case?

A) Amoicillin
B) Pregabalin
C) Cephalexin
D) Methotrexate

A 3-year-old boy presents with diarrhea. His weight is 13 Kilograms. Which of the following is the most appropriate maintenance fluid requirement for this patient?

A) 1000 ml
B) 1050 ml
C) 1150 ml
D) 1250 m

Which of the following is the mechanism of action of loperamide antidiarrheal effect?

A. Binding bile acids

B. Blocking of cholinergic receptors

C. Activation of the opioid receptors

D. Activation of alpha-2 adrenergic receptors

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